MEAL-63

Backend Configuration & Local Environment Guide

Meal Connect | Version 1.3 | Angular 21 + TypeScript • Python FastAPI • PostgreSQL • Localhost / No Docker

Implementation reference for team members. Configuration and setup steps below describe the project target; commands referencing backend modules or migrations work after those files are implemented.

1. Purpose and architecture

MEAL-63 defines the required environment variables, local development services, secret-handling rules and backend setup conventions. API endpoints and payloads are defined in MEAL-62 v1.3; the database model is defined in MEAL-6; MEAL-5 and MEAL-61 define the architecture.

Angular 21 SPA (typically localhost:4200) → HTTPS/HTTP JSON REST calls → FastAPI application (typically localhost:8000) → PostgreSQL service (typically localhost:5432). The browser must never connect directly to PostgreSQL. Docker and cloud hosting are not part of the MVP.

2. Local development services

Service	Runs where / purpose	Expected check
Angular 21 CLI	frontend/; web UI and HttpClient	npm start → browser at http://localhost:4200
FastAPI + Uvicorn	backend/; API, permissions and workflows	GET /docs or /openapi.json at http://127.0.0.1:8000
PostgreSQL	local OS service; relational persistence	psql connects with restricted app account
Python virtual environment	backend/.venv; isolates backend packages	Python and pip usable within .venv
Git	repository workflow; feature branch → develop	git status; no secrets staged

Ports above are development defaults, not guaranteed on every machine. Match CORS_ORIGINS to the actual Angular origin. Run three local services independently; no Docker compose file is needed.

3. Environment variables – backend configuration contract

Create backend/.env.example containing only names and harmless placeholder values. Each developer makes an untracked backend/.env locally. Do not commit actual credentials, signed tokens, real contact details or generated signing secrets.

Variable	Purpose / safe example	Rule
APP_ENV	development	Required: choose development for localhost
APP_HOST	127.0.0.1	Default local bind; do not expose publicly
APP_PORT	8000	Default API port

DATABASE_URL	postgresql+psycopg://<db_user>:<db_password>@localhost:5432/<db_name>	Required; fill with local credentials only in .env
JWT_SECRET_KEY	<generate-a-unique-random-value-locally>	Required if using MEAL-62 JWT baseline; never commit value
JWT_ALGORITHM	HS256	Token signing algorithm; must match implementation
ACCESS_TOKEN_EXPIRE_MINUTES	30	MVP access-token lifetime; check MEAL-62 contract
CORS_ORIGINS	http://localhost:4200	Comma-separated exact frontend origins; no wildcard
LOG_LEVEL	INFO	Avoid logging passwords, tokens, connection strings

Never copy the angle-bracket placeholders into an operational deployment without replacing them locally. Configure JWT signing with a cryptographically random secret. If a team member changes a variable name, update this guide and the actual settings parser together.

4. Safe .env.example and Git ignore rules

```
# backend/.env.example — placeholders only, safe for source control
APP_ENV=development
APP_HOST=127.0.0.1
APP_PORT=8000
DATABASE_URL=postgresql+psycopg://<db_user>:<db_password>@localhost:5432/<db_name>
JWT_SECRET_KEY=<generate-locally-do-not-commit>
JWT_ALGORITHM=HS256
ACCESS_TOKEN_EXPIRE_MINUTES=30
CORS_ORIGINS=http://localhost:4200
LOG_LEVEL=INFO
```

Repository .gitignore (or backend/.gitignore) must ignore .env, .env.*, while permitting .env.example, as well as .venv/, __pycache__/, *.pyc, test caches, database dumps and private log files. Existing frontend/.gitignore continues to exclude node_modules/ and Angular build/cache outputs.

```
# core Git exclusions (merge with existing .gitignore)
.env
.env.*
!.env.example
.venv/
__pycache__/
*.pyc
*.sql.gz
*.dump
*.log
```

5. Suggested FastAPI backend structure

```
backend/
  app/
    main.py          # FastAPI app and router registration
    core/config.py     # typed environment settings
    core/security.py   # token/hash utilities
```

```
  api/            # /api/v1 route modules
  schemas/          # Pydantic request/response models
  models/           # persistence models
  services/         # authorization and state transitions
  db/             # engine, sessions, transactions
tests/
migrations/         # Alembic migrations when added
requirements.txt    # reproducible backend dependency pins
.env.example        # safe template; commit this only
```

This is a recommended target layout, not a statement that every backend file already exists. Keep naming consistent with MEAL-62 and avoid creating competing API prefixes.

6. First-time Windows localhost setup

Commands below assume an actual backend/app/main.py entry point and committed requirements.txt. Until those files exist, run the preflight checks and create the environment; do not treat omitted modules as installation failures.

```
# From repository root
node --version
npm --version
ng version
python --version
psql --version

# Frontend (already initialized in MEAL-161)
cd frontend
npm ci
npm start

# In a separate terminal, from repository root
cd backend
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
Copy-Item .env.example .env
# Edit .env locally with your own database credentials and signing key
.\.venv\Scripts\python.exe -m uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
```

Create the local database and limited-privilege application account outside the repository; do not publish SQL commands containing actual passwords. Apply migration commands only when the team has added Alembic configuration and migration files. Exact Python, PostgreSQL, Node/npm and dependency versions are recorded in MEAL-163 and lockfiles.

7. CORS, JWT and application security

CORS allow-list must contain only the exact Angular development origin(s). A browser origin is scheme + hostname + port: localhost:4200 and 127.0.0.1:4200 are distinct. Avoid allow_origins=["*"] for authenticated applications. Angular route guards are not a security boundary; FastAPI must check authenticated user, global role, active organization membership, verification status and resource ownership on every protected action.

MEAL-62 v1.3 defines bearer JWT as the target contract. Enforce token signature and expiry on the server; keep passwords hashed (never plaintext); never put JWT signing secrets in Angular environment files or frontend bundle. Restrict database privileges to the minimum needed by the application and use a separate migration role if feasible.

8. Data integrity and API behaviour

MEAL-6 is the source of truth for tables, enums and constraints. Restaurant/bakery listings require approved organizations. Only approved NGOs request quantities from existing donations. Restaurants approve the entire requested_quantity or reject; there is no partial approval or approved_quantity field. A PostgreSQL transaction must serialize competing approvals and update allocated_quantity atomically. Backend rejects expired/cancelled donations, invalid transitions and unauthorized actors; audit status changes in the same transaction.

MEAL-62 defines response and error shapes (including 401, 403, 404, 409 and 422), pagination, filtering and endpoint names. FastAPI /docs reflects implemented code; when code and contract differ, correct the implementation or deliberately version the contract before Angular integration.

9. Practical startup checks and troubleshooting

Check	Expected result / action
Angular loads	Browser opens localhost:4200; npm dependencies installed
FastAPI starts	Uvicorn binds on configured host:port; /docs available
Database connection	App connects through DATABASE_URL; no credentials printed
CORS	Angular HttpClient request succeeds; allow correct exact origin
Validation	Invalid body returns MEAL-62 standard 422 structure
Auth and access	Unauthenticated call denied 401; forbidden scoped action denied 403
Quantity concurrency	Two approvals cannot reserve beyond total; losing request receives 409
Secret hygiene	git status shows no .env; no secrets in code, examples, JIRA or screenshots

10. Reference boundaries

MEAL-5 v1.3: architecture and workflow. MEAL-61 v1.3: component responsibilities/data flow. MEAL-6: PostgreSQL logical model, constraints and status vocabulary. MEAL-62 v1.3: canonical HTTP endpoint and payload contract. MEAL-163: development tool versions and prerequisites. MEAL-63: authoritative local backend configuration and environment guidance.

Changes to endpoint schemas belong first in MEAL-62; changes to database constraints belong in MEAL-6 and migrations; environment variable names and service setup belong in MEAL-63. This separation keeps the implementation guidance consistent.